

Random Variable

The bridge between random outcomes and mathematics

A random variable is a function that maps the outcomes of a random experiment to **numerical values**.



FORMAL DEFINITION

Let S be the sample space (set of all possible outcomes).
A random variable X is a function $X : S \rightarrow \mathbb{R}$.

THE INTUITION



We often care about **qualitative outcomes** (e.g., Heads or Tails).



Machines and mathematical models operate on numbers.



A random variable defines the “translation” of **qualitative outcomes** into **quantitative data**.

HOW IT WORKS: A MAPPING FROM OUTCOMES TO NUMBERS

Sample Space (S)

Outcomes

ω_1

ω_2

$\vdots$

ω_n

Random Variable

$X : S \rightarrow \mathbb{R}$

$X(\omega_1)$

$X(\omega_2)$

$\vdots$

$X(\omega_n)$

Numerical Values

Real Numbers

x_1

x_2

$\vdots$

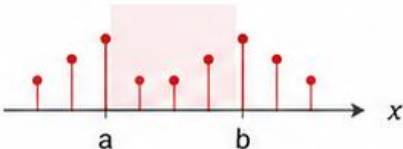
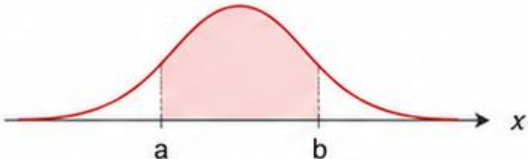
x_n

EXAMPLE: COIN TOSS

Outcome (ω)	Random Variable (X)	Value
Heads (H)	Let $X = 1$ if Heads, 0 if Tails	1
Tails (T)		0

2 Taxonomy of Random Variables

The nature of the numerical mapping determines the mathematical framework required for analysis.

FEATURE	 DISCRETE RANDOM VARIABLE	 CONTINUOUS RANDOM VARIABLE
 Values	Countable (gaps between values) 	Uncountable (any value in an interval) 
 Primary Tool	Probability Mass Function (PMF) $p_X(x) = P(X = x)$	Probability Density Function (PDF) $f_X(x)$
 Logic	$P(X = x)$ is the probability of exact occurrence	$P(X = x)$ is essentially zero
 Probability	Summation of discrete points $P(a \leq X \leq b) = \sum_{x=a}^b P(X = x)$ 	Area under the curve (Integral) $P(a \leq X \leq b) = \int_a^b f_X(x) dx$ 



Key Takeaway: Discrete variables deal with **counts** and use **sums**. Continuous variables deal with **measures** and use **integrals**.

3 Probability Distributions

We describe uncertainty using functions that assign probabilities to outcomes.

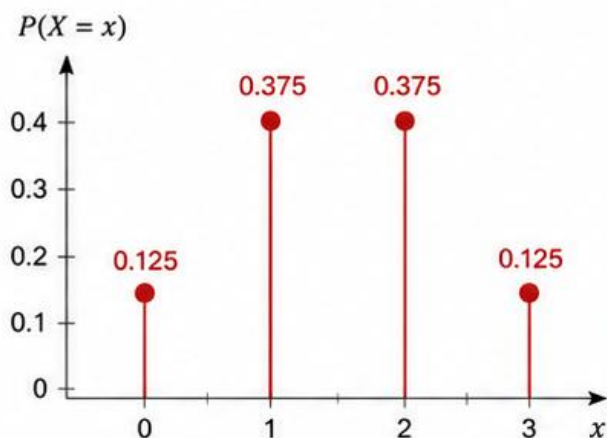
3.1 PROBABILITY MASS FUNCTION (PMF)

For **discrete** variables, the PMF assigns a specific probability to each individual value.



What it looks like

Example: Number of heads in 3 coin tosses (X)



Key Rules

1

Summation Rule

The sum of all probabilities must equal 1:

$$\sum_i P(X = x_i) = 1$$

2

Non-negativity

For all x :

$$0 \leq P(X = x) \leq 1$$



Intuition

The PMF places “probability mass” at specific points. The total mass adds up to 1.

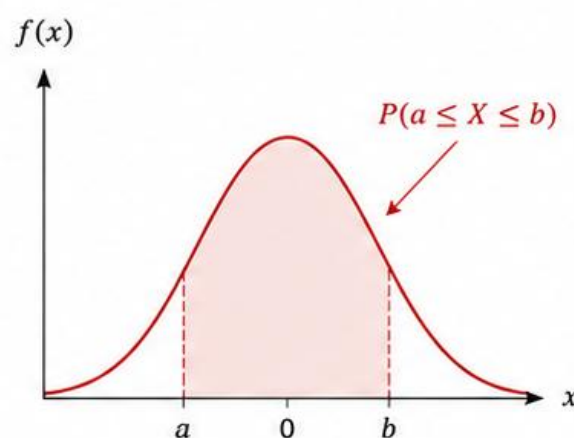
3.2 PROBABILITY DENSITY FUNCTION (PDF)

For **continuous** variables, we cannot assign a non-zero probability to a single point. Instead, we describe the **density** of probability.



What it looks like

Example: Normal distribution $X \sim \mathcal{N}(0, 1)$



Key Rules

1

Range Probability

Probability that X falls in an interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

2

Normalization Rule

The total area under the PDF curve must equal 1:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



Intuition

The PDF describes how probability is spread over an interval. The total area under the curve equals 1.

1 Sampling & Representation

Sampling is the foundation of inference. If the sample is biased, the inference will be invalid, **regardless of the statistical rigor applied later.**

1.1 TAXONOMY OF SAMPLING

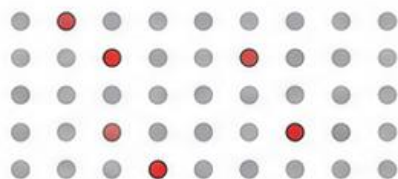


PROBABILITY SAMPLING

Each element has a known non-zero probability of selection.
Necessary for valid inference.

1 Simple Random Sampling (SRS)

Every combination of n elements has an equal probability.

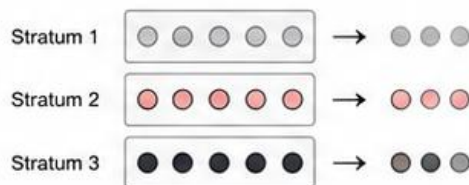


○ Population ● Sample

Example: Randomly selecting 50 students from a university student list.

2 Stratified Sampling

Ensures representation of subgroups (strata) by sampling from each independently.

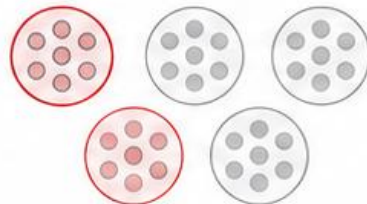


○ Population ● Sample

Example: Sampling employees from each department (HR, Sales, Engineering) proportionally.

3 Cluster Sampling

Efficient for large geographical areas; samples whole groups (clusters).



○ Population ● Selected cluster

Example: Randomly selecting schools in a city, then surveying all students in those schools.



NON-PROBABILITY SAMPLING

Convenience sampling; lacks theoretical justification for generalizing results.



Convenience Sampling

Selects the most accessible individuals.

Example: Surveying people in a mall.



Judgment/Purposive Sampling

Relies on researcher judgment.

Example: Experts selected for an interview.



Quota Sampling

Fills pre-set quotas for key characteristics.

Example: Gender quota in street interviews.



Snowball Sampling

Participants recruit other participants.

Example: Research on hidden populations.



KEY TAKEAWAY

Use probability sampling methods whenever possible to ensure **representativeness** and enable **valid statistical inference**.

2 Point Estimation

A point estimator is a **rule (statistic)** used to estimate a population parameter using sample data.



THE IDEA

We use a single number computed from the sample to “point” to the true (unknown) parameter.

EXAMPLE

Estimate the population mean μ using the sample mean $\bar{X}$.



COMMON POINT ESTIMATORS

Parameter	Point Estimator ($\hat{\theta}$)	Distribution (under i.i.d.)
Mean (μ)	$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$	$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$
Proportion (p)	$\hat{p} = \frac{X}{n}$	$E[\hat{p}] = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}$
Variance (σ^2)	$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$	$E[S^2] = \sigma^2$



KEY TAKEAWAY

A good point estimator is unbiased (hits the target on average) and efficient (as precise as possible).



CRITERIA FOR “GOOD” ESTIMATORS

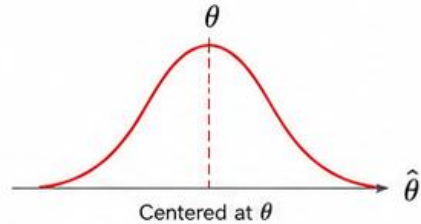
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UNBIASEDNESS

The sampling distribution of the estimator is centered at the true parameter.

$$E[\hat{\theta}] = \theta$$



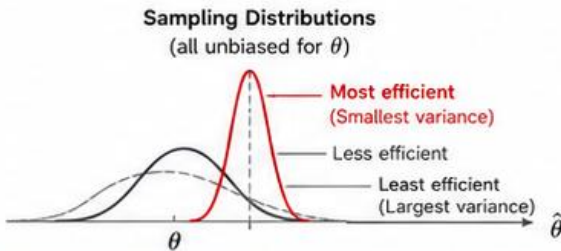
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EFFICIENCY

Among all unbiased estimators, it has the smallest variance.

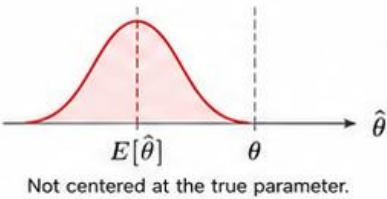
$$\text{Var}(\hat{\theta}) \text{ is minimum}$$



VISUAL INTUITION: WHAT MAKES AN ESTIMATOR “GOOD”?

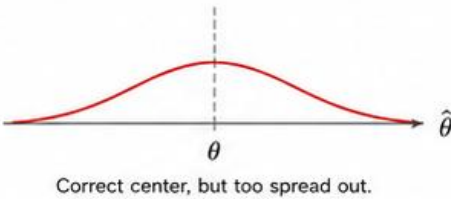
Biased Estimator

$$E[\hat{\theta}] \neq \theta$$



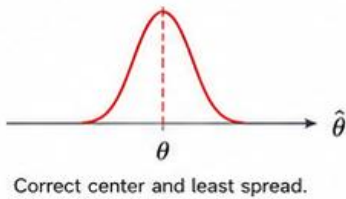
Unbiased but Not Efficient

Variance is large



Unbiased and Efficient

Minimum variance



WHY IT MATTERS

Good estimators yield more reliable inference, narrower intervals, and better decisions.



EXAMPLE IN PRACTICE

For normally distributed data, $\bar{X}$ is unbiased and efficient for estimating μ .

3

The Central Limit Theorem (CLT)

The CLT is the **bridge** between raw data and the Normal Distribution. It states that for a sufficiently large sample size ($n \geq 30$), the sampling distribution of the sample mean $\bar{x}$ will be **approximately Normal**, regardless of the population distribution shape.

THE BIG IDEA



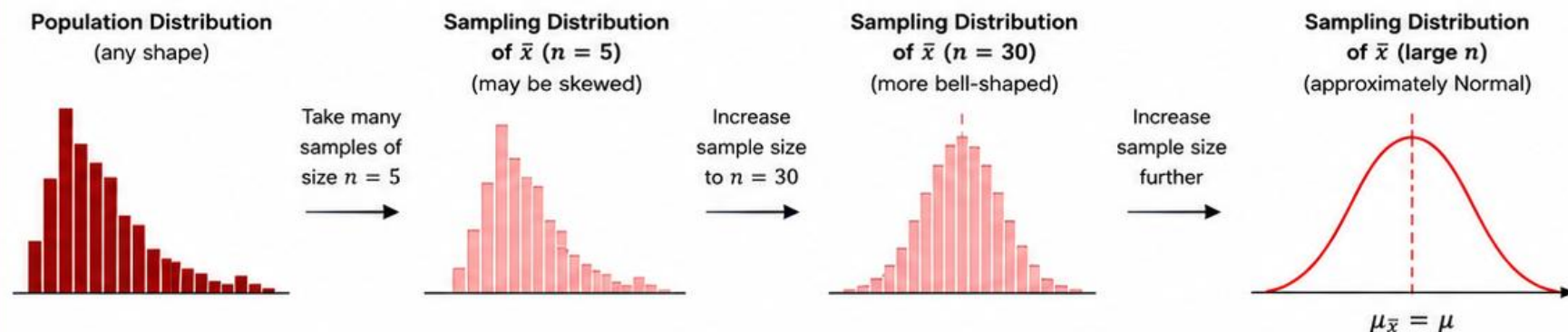
No matter what the population looks like, the distribution of sample means looks approximately Normal when n is large.

INTUITION



Averaging many observations smooths out irregularities. The more data we average, the more "bell-shaped" the sampling distribution becomes.

FROM ANY POPULATION TO (APPROX.) NORMAL SAMPLING DISTRIBUTION



Rule of Thumb: When $n \geq 30$, the sampling distribution of $\bar{x}$ is approximately Normal.

PARAMETERS OF THE SAMPLING DISTRIBUTION OF $\bar{x}$

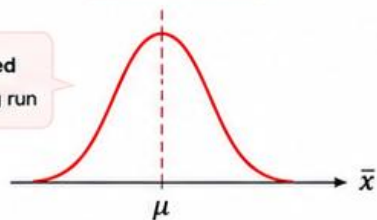


1 CENTER (MEAN)

The mean of the sampling distribution of $\bar{x}$ equals the population mean.

$$\mu_{\bar{x}} = \mu$$

Unbiased
in the long run

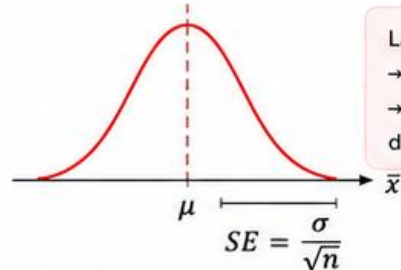


2 SPREAD (STANDARD ERROR)

The spread of the sampling distribution is the Standard Error (SE).

$$SE = \frac{\sigma}{\sqrt{n}}$$

Larger n
→ smaller SE
→ narrower distribution



KEY IMPLICATIONS



Foundation for Inference

Allows us to use Normal-based methods (confidence intervals, hypothesis tests) even when the population is not Normal, as long as n is large.



Precision Increases with Sample Size

As n increases, $SE = \frac{\sigma}{\sqrt{n}}$ decreases, so sample means cluster more tightly around μ .



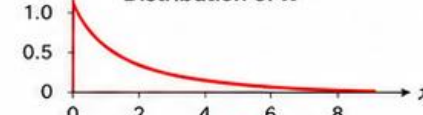
Why "30"?

$n = 30$ is a rule of thumb from empirical evidence; many distributions look sufficiently Normal by then (even skewed ones).

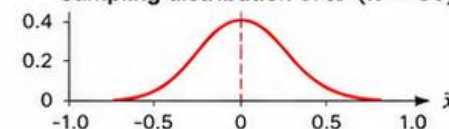
EXAMPLE

Population: Exponential ($\lambda = 1$)
(highly right-skewed)

Distribution of X



Sampling distribution of $\bar{X}$ ($n = 30$)



With $n = 30$, the distribution of $\bar{X}$ is approximately Normal.

4 Interval Estimation

Point estimates are inherently imprecise. **Confidence Intervals (CI)** quantify this uncertainty by providing a range of plausible values for the parameter.

4.1 CONFIDENCE INTERVAL STRUCTURE

A confidence interval gives a range of plausible values for a population parameter θ .

$$\text{CI} = \text{Point Estimate} \pm (\text{Critical Value} \times \text{Standard Error})$$



1. Point Estimate

Best single guess of the parameter.

Example: $\bar{x}$ estimates μ .



2. Critical Value

Determined by the confidence level $(1 - \alpha)$ and the chosen distribution.

Example: z^* or t^* .



3. Standard Error (SE)

Measures the variability of the estimator across samples.

Example: $\frac{\sigma}{\sqrt{n}}$ or $\frac{s}{\sqrt{n}}$.

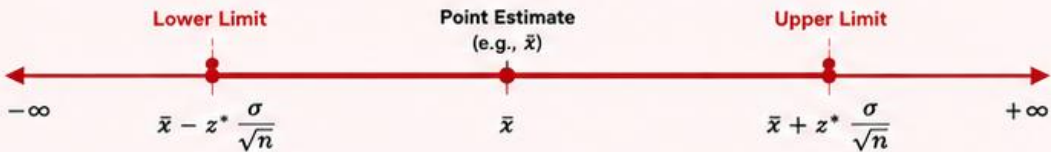


4. Interval

Range of values that is expected to contain θ in $100(1 - \alpha)\%$ of repeated samples.

Visual Interpretation

We are $100(1 - \alpha)\%$ confident that the true parameter θ lies within this interval.



Z vs. t: KEY DIFFERENCES

Feature	Z-Interval (σ Known)	T-Interval (σ Unknown)
Underlying distribution	Standard Normal $N(0, 1)$	t -Distribution with $df = n - 1$
Critical values	Fixed (do not depend on n)	Depend on n (via df)
Tails	Lighter	Fatter (more conservative)
When to use	σ known or n large (≥ 30)	σ unknown and n small/moderate
Consequence	Narrower intervals	Wider intervals (more honest)



KEY TAKEAWAY

Use **Z-intervals** when σ is known or n is large.

Use **T-intervals** when σ is unknown, especially for small n .

4.2 SELECTION OF DISTRIBUTION

Z-INTERVAL (σ KNOWN)



- Used when the population standard deviation σ is known.
- Uses Z -scores from the Standard Normal distribution $N(0, 1)$.
- Appropriate for large samples ($n \geq 30$) even if σ is estimated.

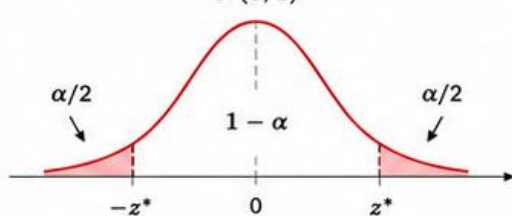
CI for Mean (σ Known)

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

where z^* is the critical value from $N(0, 1)$.

Standard Normal (Z) Distribution

$N(0, 1)$



Examples of z^* :

Confidence Level $(1 - \alpha)$	90%	95%	99%
z^*	1.645	1.960	2.576

T-INTERVAL (σ UNKNOWN)



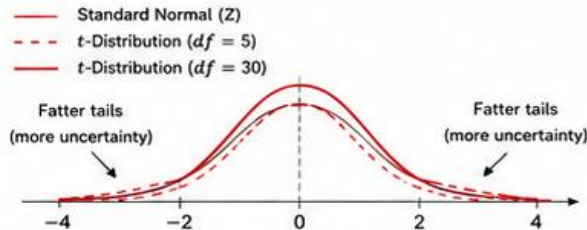
- Used when the population standard deviation σ is unknown (estimated by s).
- Uses the t -distribution with $df = n - 1$.
- Has "fatter tails" than the normal distribution, especially for small n , reflecting extra uncertainty.

CI for Mean (σ Unknown)

$$\bar{x} \pm t^*_{(df)} \frac{s}{\sqrt{n}}$$

where $t^*_{(df)}$ is the critical value from the t -distribution with $df = n - 1$.

t-Distribution vs. Standard Normal



Examples of $t^*_{(df)}$ (two-tailed):

Confidence Level $(1 - \alpha)$ $df = n - 1$	90%	95%	99%
5	2.015	2.571	4.032
10	1.812	2.228	3.169
30	1.697	2.042	2.750
∞ (Z)	1.645	1.960	2.576



IMPORTANT NOTE

The confidence level $(1 - \alpha)$ is the long-run success rate of the method, not the probability that a specific interval contains θ .



EXAMPLE

A 95% CI procedure means that in 95% of repeated samples, the interval constructed will contain the true parameter.

Parametric vs. Non-Parametric Methods

Choosing between model-based precision and assumption-free robustness



Core Decision: Do we assume a specific population distribution (parametric) to gain power, or do we minimize assumptions (non-parametric) to gain robustness?

At a Glance Comparison

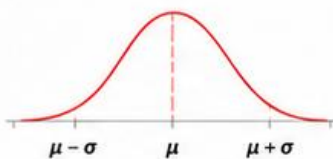
	Parametric	Non-Parametric
Assumptions	Strong (Distributional form)	Minimal (Distribution-free)
Statistical Power	High (if assumptions are met)	Lower (but stable)
Robustness	Fragile to violations	Robust to violations
Interpretability	Parameter estimates (e.g., μ , σ , β)	Ranks, medians, proportions



1. Parametric Methods: The “Model-Based” Approach

Core Assumption

Population follows a specific mathematical form.



Typically Normal (Gaussian)

Mechanism

Inference focuses on estimating a **small set of parameters**.

Normal Distribution

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Parameters: μ, σ

Examples



Student's t-test



ANOVA



Linear Regression

Computational Trade-offs



Statistical Power

When assumptions hold (e.g., Normality), parametric tests provide the highest possible power.



Fragility

If the population is skewed or heavy-tailed, parametric tests can produce biased estimates and misleading p-values.



2. Non-Parametric Methods: The “Distribution-Free” Approach

Core Assumption

No specific distributional form is assumed.



Works for any distribution (skewed, heavy-tailed, multimodal)

Mechanism

Inference is based on **ranks, signs, or counts**.

Rank Transformation

$$x_1, x_2, \dots, x_n$$



$$R_1, R_2, \dots, R_n$$

Use ranks instead of raw values

Examples



Mann–Whitney U Test (Wilcoxon Rank-Sum)



Kruskal–Wallis Test



Wilcoxon Signed-Rank Test

Computational Trade-offs



Robustness

Performs reliably across a wide range of distributions, outliers, and skewed data.



Lower Power

Because it uses less information (ranks instead of exact values), power is lower when parametric assumptions are true.

6. Common Examples



Comparing Two Independent Groups
t-test vs. Mann–Whitney U



Comparing >2 Groups
ANOVA vs. Kruskal–Wallis



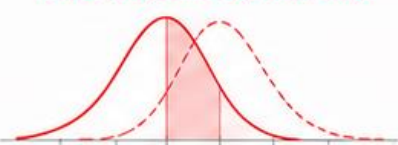
Relationship Between Variables
Linear Regression vs. Spearman Correlation



Paired / Repeated Measurements
Paired t-test vs. Wilcoxon Signed-Rank

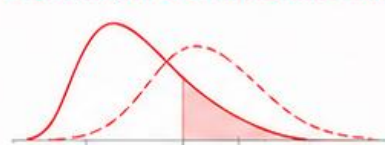
3. Visual Intuition

When Assumptions Hold (Normal Data)



Parametric = More Power (narrower uncertainty)

When Assumptions Are Violated (Skewed Data)



Non-Parametric = More Reliable (wider but safer)

4. Decision Guide



5. Key Takeaway



Parametric methods bet on a model to “win big” on **power**.

Non-parametric methods avoid the bet to “stay safe” on **robustness**.

Choose based on your **confidence** in the assumptions and the **cost of being wrong**.



Practical Rule of Thumb: Start with exploratory analysis. If data looks approximately Normal and sample size is moderate/large → parametric.

If data is skewed, ordinal, or contains outliers → non-parametric.

Inferential Statistics: Principles of Inference, Sampling & Parameter Estimation

From Sample Data to Population Truth



Core Question: How do we use data from a subset (**sample**) to make rigorous, quantifiable conclusions about a larger group (**population**)?

The Inferential Pipeline

1. Population



All individuals or observations of interest

Parameter: θ

2. Sampling



Select a subset (sample) of size n using a sampling design

Goal: Representativeness

3. Data



Observe data
 $X_1, X_2, \dots, X_n$

4. Estimation



Compute a point estimate $\hat{\theta}$ of the parameter θ

Example: $\bar{x}$ for μ

5. Inference



Quantify uncertainty, test hypotheses, and make conclusions about θ

Using probability!

Population vs. Sample

Population



- Size: N (very large)
- Parameter: θ (fixed)
- Example: True mean μ

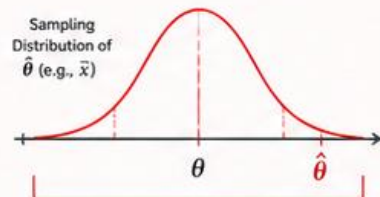
Sample



- Size: n ($\ll N$)
- Statistic: $\hat{\theta}$ (random)
- Example: Sample mean $\bar{x}$

Sampling

Variability of Estimates



Sampling Variability

Different samples $\rightarrow$ different $\hat{\theta}$
centered (ideally) around θ

Sources of Sampling Error



Random Chance

Natural variability due to chance in selection and observation



Sampling Design

Poor sampling methods can introduce bias and increase error



Non-response

Systematic differences between respondents and non-respondents



Measurement Error

Inaccuracies in data collection or instruments

The Fundamental Tension



Desire

We want to know the
Population Parameter (θ)

e.g., true mean height (μ)
of the Indian population



Feasibility Gap

Observing the entire
population is generally
computationally,
temporally, and
financially infeasible



Statistical Solution

We use a **Sample**
($X_1, X_2, \dots, X_n$) to derive

a **Point Estimate ($\hat{\theta}$)**
of the parameter

Core Concepts at a Glance



Parameter (θ)

A fixed (usually unknown) numerical summary
of a population (e.g., μ , σ^2 , p).



Statistic

A numerical summary computed from sample
data (e.g., $\bar{x}$, s^2 , $\hat{p}$).



Estimator ($\hat{\theta}$)

A rule or method used to calculate an estimate
of θ from the sample.



Point Estimate

A single value $\hat{\theta}$ used to estimate the parameter θ
(e.g., $\bar{x}$ is the point estimate of μ).



Uncertainty

Quantified using measures like Standard Error,
Confidence Intervals, and p -values.



Inferential statistics provides the mathematical bridge
from sample to population.

- We estimate the unknown parameter θ using $\hat{\theta}$.
- We quantify the uncertainty in $\hat{\theta}$.
- We make decisions or draw conclusions with a controlled risk of error.

“ Statistics is the art of learning from data
about a world too big to measure completely. ”



Why It Matters



Enables data-driven
decision making



Generalizes findings
beyond the sample



Quantifies confidence
in conclusions



Controls risk of making
wrong decisions



Foundation of machine learning,
A/B testing, surveys, and more

2 Sampling Theory and Representation

The quality of our inference is determined by the quality of our sample.



Core Principle: Inference is only as valid as the sampling procedure that precedes it.

Concept	Description	Engineering / ML Context
 Simple Random Sampling	Each member of the population has an equal and independent probability of selection.	 Basis for most i.i.d. assumptions in machine learning models.
 Stratified Sampling	Population is divided into homogeneous subgroups (strata). A random sample is drawn from each stratum.	 Essential when dealing with minority classes or imbalanced datasets.
 Sampling Error	The difference between a sample statistic and the true population parameter due to random sampling.	 Quantified by Standard Error (SE). Drives confidence in estimates.



Computational Note: As sample size n approaches population size N , sampling error approaches zero.



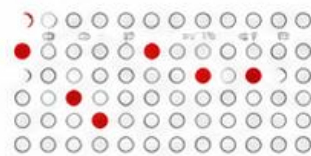
$$\lim_{n \rightarrow N} SE(\hat{\theta}) = 0$$



Key Takeaway: Better sampling $\rightarrow$ Lower sampling error $\rightarrow$ More precise and reliable inference.

Common Sampling Procedures

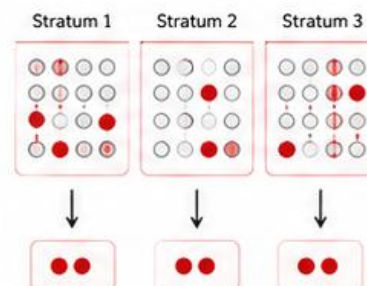
Simple Random Sampling



○ Population (N) ● Sample (n)

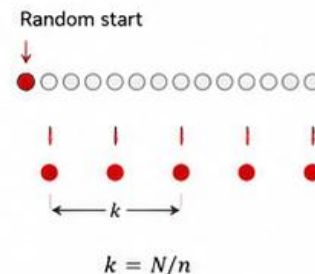
Every unit has equal chance of being selected.

Stratified Sampling



Ensures representation from each important subgroup.

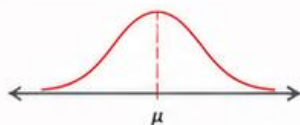
Systematic Sampling



Select every k -th unit after a random starting point.

Sampling Error and Standard Error (SE)

Population mean: μ
Sample mean: $\bar{X}$
Sampling error: $\bar{X} - \mu$



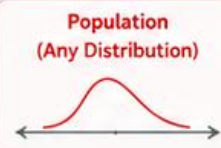
Larger $n \rightarrow$ Smaller SE $\rightarrow$ Less sampling error

Standard Error of the Mean

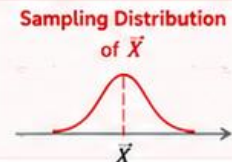
$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

σ = population standard deviation
 n = sample size
SE decreases as n increases.

Central Limit Theorem (CLT): Why We Get Normality



Take all possible samples of size n



As n increases



CLT Statement

As $n \rightarrow \infty$, the sampling distribution of the mean $\bar{X}$ approaches a normal distribution, regardless of the shape of the population distribution, provided $E[X^2] < \infty$.

Implication

Enables valid confidence intervals and hypothesis tests for $\bar{X}$ when n is sufficiently large.



Practical Insight: Good sampling design is as important as advanced statistical methods. Garbage in $\rightarrow$ Garbage out.



Rule of Thumb: For the CLT, $n \geq 30$ is often sufficient for approximate normality.

Parameter Estimation: Theory to Practice

From a single number to a quantified range of uncertainty.



Goal: Infer the true population parameter θ using sample data $X_1, X_2, \dots, X_n$.

3.1 Point Estimation

A point estimate is a single numerical value used to approximate a population parameter θ .

Method of Moments (MoM)



Equate sample moments to population moments and solve for θ .

Example (Normal $N(\mu, \sigma^2)$)

$$E[X] = \mu \Rightarrow \bar{X} = \mu$$

MoM estimator of μ : $\hat{\mu}_{MoM} = \bar{X}$

Maximum Likelihood Estimation (MLE)



Find θ that maximizes the likelihood of observing the data.

Example (Normal $N(\mu, \sigma^2)$, σ^2 known)

$$\mathcal{L}(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

MLE of μ : $\hat{\mu}_{MLE} = \bar{X}$

Bias vs. Variance Trade-off



Unbiased if $E[\hat{\theta}] = \theta$.
In practice, we may accept small bias to reduce variance (lower MSE) and overfitting.

Mean Squared Error (MSE)

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Properties of a Good Estimator



Unbiased
 $E[\hat{\theta}] = \theta$



Low Variance
 $\text{Var}(\hat{\theta})$ is small



Consistent
 $\hat{\theta} \xrightarrow{P} \theta$ as $n \rightarrow \infty$

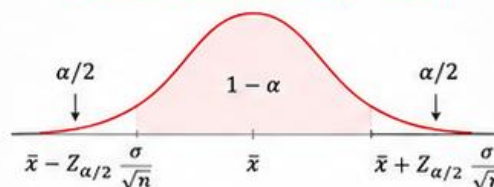


Efficient
Achieves lowest variance among unbiased estimators

3.2 Interval Estimation (Confidence Intervals)

Instead of a "spear" (point estimate), we use a "net" (confidence interval) to capture uncertainty.

Population Variance Known (σ^2 known)

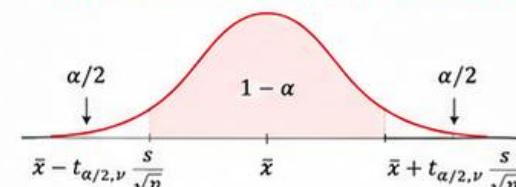


Confidence Interval (CI)

$$CI = \bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

where $Z_{\alpha/2}$ is the standard normal critical value.

Population Variance Unknown (σ^2 unknown)



Confidence Interval (CI)

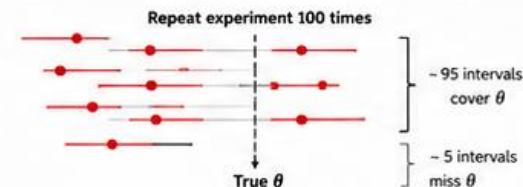
$$CI = \bar{x} \pm t_{\alpha/2, \nu} \left(\frac{s}{\sqrt{n}} \right)$$

where $t_{\alpha/2, \nu}$ is the t critical value, $\nu = n - 1$.



The Frequentist Trap (Interpretation)

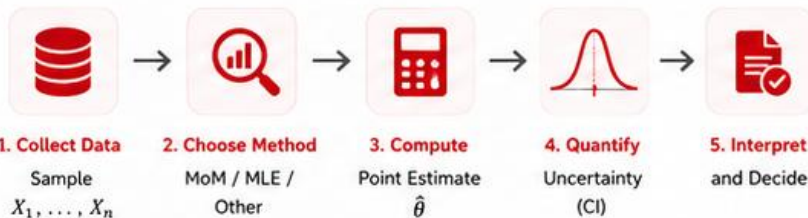
A 95% Confidence Interval does NOT mean there is a 95% probability the true parameter is inside. It means that if we repeated the experiment 100 times, 95 of the intervals constructed would contain the true parameter.



Estimator Comparison

Estimator	Bias	Variance	Consistency	Efficiency	Notes
Sample Mean $\bar{X}$	0	σ^2/n	✓	✓	Best linear unbiased estimator (BLUE)
MLE (Normal μ)	0	σ^2/n	✓	✓	Efficient (achieves Cramér-Rao bound)
MoM (Normal μ)	0	σ^2/n	✓	✓	Same as MLE in this case
Biased Estimator	$\neq 0$	Lower possible	✓	✗	May reduce MSE in some settings

Estimation Workflow



Key Quantities

Sample Mean	$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
Sample Std. Dev.	$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$
Standard Error (Mean)	$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ (known σ), $\frac{s}{\sqrt{n}}$ (unknown σ)
Critical Values	$Z_{\alpha/2}$ (Normal), $t_{\alpha/2, \nu}$ (t-distribution)



Key Takeaways



Sampling quality drives inference quality.



Point estimates give a best guess.



Confidence intervals quantify uncertainty.



Larger $n \rightarrow$ smaller SE $\rightarrow$ narrower intervals.



Understand assumptions behind every method.

4 The Hypothesis Testing Framework

A formal procedure for deciding between two mutually exclusive claims.



Goal: Use sample data to decide whether to reject the Null Hypothesis (H_0) in favor of the Alternative (H_1).

1 Null Hypothesis (H_0)



The status quo.
"No effect" or
"No difference".

Assumed true unless
sufficient evidence
suggests otherwise.

Example:

$$H_0: \mu = \mu_0$$

2 Alternative Hypothesis (H_1)



The claim we wish
to provide evidence for.

Can be one-sided
($>$ or $<$) or two-sided
($\neq$).

Example:

$$H_1: \mu \neq \mu_0$$

3 Significance Level (α)



Risk tolerance for making
a Type I Error
(False Positive).

Common choices:
0.10, 0.05, 0.01

α defines the size of the
rejection region.

4 Test Statistic

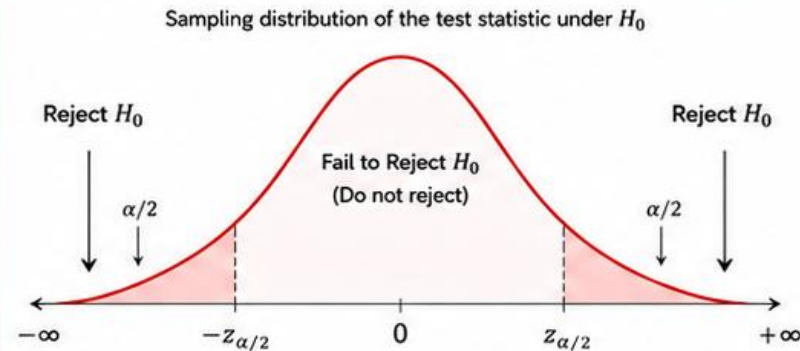


A signal-to-noise
ratio that summarizes
the evidence in the data.

$$\text{Test Statistic} = \frac{\text{Signal}}{\text{Noise}}$$

Examples: z , t , χ^2 , F

Visualizing Decision Regions (Example: Two-Sided Test)



The Testing Procedure: Step-by-Step



1. State Hypotheses

Formulate H_0 and H_1 .



2. Choose α

Set the significance level (e.g., 0.05).



3. Select Test Statistic

Choose the appropriate test statistic based
on data and assumptions.



4. Determine Distribution Under H_0

Know the sampling distribution of the test
statistic assuming H_0 is true.



5. Compute Test Statistic

Calculate the test statistic from the sample data.



6. Make Decision

If statistic falls in rejection region $\rightarrow$ Reject H_0 .
Otherwise $\rightarrow$ Fail to reject H_0 .



7. Interpret in Context

Translate the statistical decision into the
problem domain.

Types of Errors

	Truth	
	H_0 is True	H_0 is False (H_1 is True)
Decision	Reject H_0	Type I Error (False Positive) Probability = α
	Fail to Reject H_0	Correct Decision Probability = $1 - \beta$



Type I Error (α): Rejecting H_0 when it is actually true.
Consequence: False alarm.



Type II Error (β): Failing to reject H_0 when H_1 is true.
Consequence: Missed detection.



Power ($1 - \beta$): Probability of correctly rejecting H_0 when H_1 is true.
Higher power is better.

Common Test Statistics

Test	Parameter	Test Statistic	Use Case
Z-test	Mean (μ) (σ known)	$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	Large n or population σ known
t-test	Mean (μ) (σ unknown)	$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$ ($\nu = n - 1$)	Small/medium n , σ unknown
χ^2 -test	Variance (σ^2) or Goodness of Fit	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ (ν df)	Categorical data, variances
F-test	Variance Ratio (σ_1^2 / σ_2^2)	$F = \frac{s_1^2}{s_2^2}$ (ν_1, ν_2 df)	Compare two variances



**Key
Takeaways**



H_0 is the default
assumption.



α controls the risk of
False Positives (Type I).



β controls the risk of
False Negatives (Type II).



The test statistic measures
how extreme the data is
under H_0 .



Small p -value ($\leq \alpha$)
 $\rightarrow$ Reject H_0 .



Always interpret results in
the context of the problem
and consequences.

p-value Connection

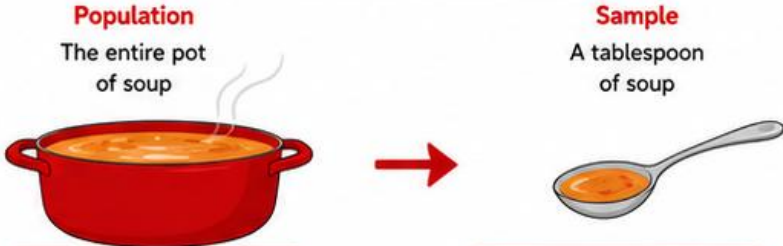
p -value = $P(\text{data as or more extreme than observed} \mid H_0 \text{ is true})$

Decision: Reject H_0 if p -value $\leq \alpha$.

1 The Art of Sampling

Sampling: The foundation of inferential statistics

The Soup Analogy



Why not analyze the entire population?

- ⌚ Very time-consuming
- ₹ Costly and often infeasible

💡 We use a small sample to make conclusions about the larger population.

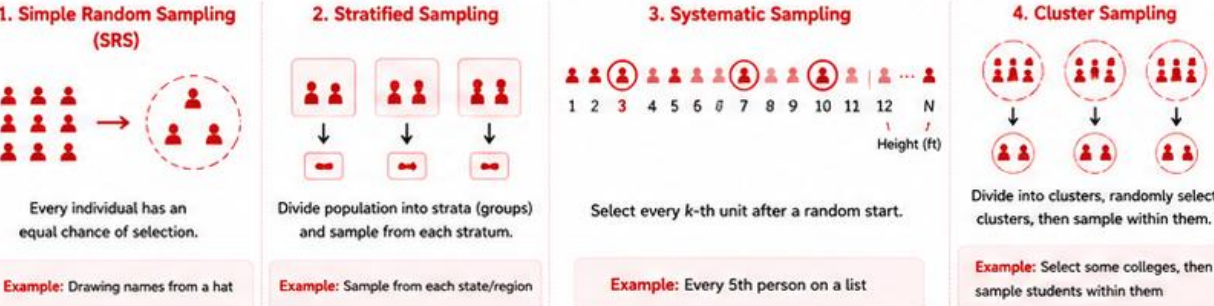
Parameter: How good is the soup?

Estimate: Taste and decide how good the soup is

Real-World Examples



Types of Probability Sampling



★ **Golden Rule:** The quality of your conclusions is only as good as the quality of your sample.



Population (N)

The complete set of all individuals or objects of interest.
e.g., All university graduates in India



Sample (n)

A subset of the population selected for analysis.
e.g., 1,000 graduates from one university



Our goal: Use the sample to estimate the population parameter as accurately as possible.

Two Broad Sampling Mechanisms



Probability Sampling

Every unit has a known, non-zero chance of being selected.

- ✓ Adds randomness
- ✓ Allows statistical inference
- ✓ Less bias, more reliable



Non-Probability Sampling

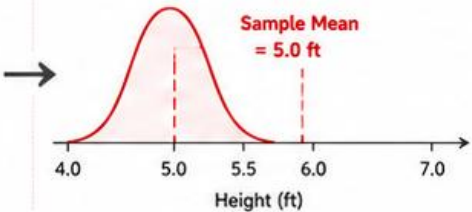
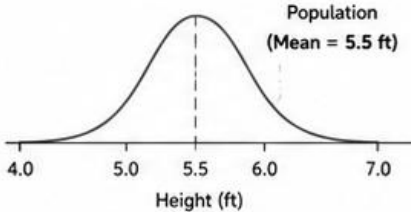
Selection is based on convenience or subjective judgment.

- ✓ Easier and cheaper
- ✓ Prone to bias
- ✓ Limited generalizability

Why Non-Probability Sampling Can Be Biased

Goal: Estimate average height of people in India

Biased Sample (from one short-height region)



Consequence

Our estimate systematically deviates from the truth.

Key Takeaways



A sample is a subset of the population.



Probability sampling supports valid statistical inference.



Non-probability sampling is convenient but can be biased.



Good sampling design leads to reliable conclusions.



Representativeness is the key to inference quality.



Remember: Randomness reduces bias. Representation increases reliability.

3.1 Point Estimation: Theory and Properties

Using a sample statistic to approximate an unknown population parameter with a single value.

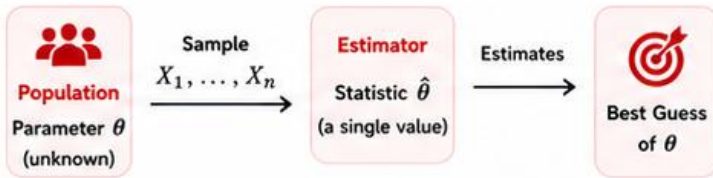


Goal

Obtain the "best guess" $\hat{\theta}$ of a population parameter θ using sample data $X_1, X_2, \dots, X_n$.

1 What is Point Estimation?

A point estimate is a single numerical value used to estimate an unknown population parameter.



Key Idea: Different samples produce different estimates, but a good estimator targets the true parameter on average.

2 Common Point Estimates

We estimate three key population parameters using corresponding sample statistics.

Population Parameter	Used For	Point Estimate (Sample Statistic)	Formula
μ Population Mean (μ)	Continuous variables	Sample Mean ($\bar{X}$)	$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
p Population Proportion (p)	Categorical variables	Sample Proportion ($\hat{p}$)	$\hat{p} = \frac{X}{n}$ (X = # of successes)
σ^2 Population Variance (σ^2)	Measure of dispersion	Sample Variance (S^2)	$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

Note: n = sample size, X_i = i -th observation, $\bar{X}$ = sample mean

3 What Makes an Estimator "Good"?

A good estimator has desirable statistical properties.



Unbiasedness

$$E[\hat{\theta}] = \theta$$

On average, the estimator hits the true parameter.



Consistency

$$\hat{\theta} \xrightarrow{p} \theta \text{ as } n \rightarrow \infty$$

As sample size increases, the estimator converges to the true parameter.



Efficiency

Smallest variance among unbiased estimators

More precise estimates (less variability).



Sufficiency

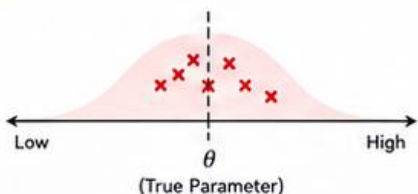
Uses all information in the sample

No other statistic based on the same sample provides more information.

4 Bias vs. Unbiasedness

Unbiased Estimator

Estimates are centered around the true parameter θ .

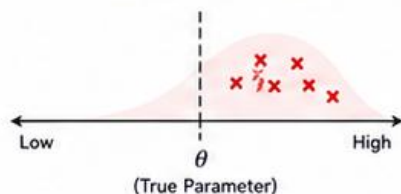


$$E[\hat{\theta}] = \theta$$

Average of estimates = True value

Biased Estimator

Estimates are systematically away from θ .



$$E[\hat{\theta}] \neq \theta$$

Average of estimates $\neq$ True value



Example: if $E[\hat{\theta}] = \theta + 2$, the estimator overestimates by 2 on average.

5 Bias-Variance Trade-off

An estimator with lower variance may have some bias, and vice versa.

Mean Squared Error (MSE)

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Low Variance
Precise



Low Bias
Accurate



Goal: Find the right balance to minimize MSE.

6 Examples of Point Estimation

Parameter θ	Estimator $\hat{\theta}$	Unbiased?	Notes
Mean (μ)	$\bar{X}$	✓	Best linear unbiased estimator (under i.i.d. assumptions)
Proportion (p)	$\hat{p} = \frac{X}{n}$	✓	$X \sim \text{Binomial}(n, p)$
Variance (σ^2)	$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$	✓	Uses $n-1$ (Bessel's correction) to be unbiased
Variance (σ^2)	$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$	✗	Biased (underestimates σ^2)



Key Takeaway

Point estimation gives a single "best guess" of a parameter.
A good estimator is unbiased, consistent, and efficient.



Remember: Any single estimate is uncertain.
We move to **interval estimation** to quantify this uncertainty.



Next Up: Interval Estimation (Confidence Intervals)
— capturing uncertainty with a range, not a point.

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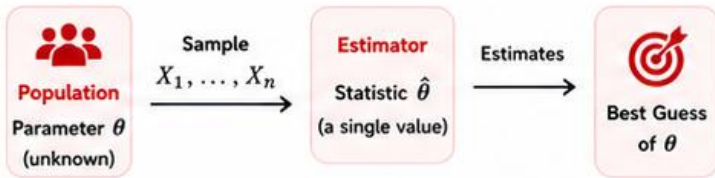


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Sufficiency

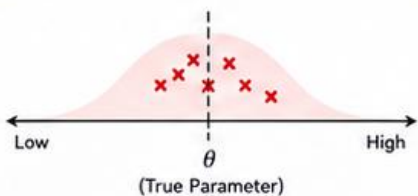
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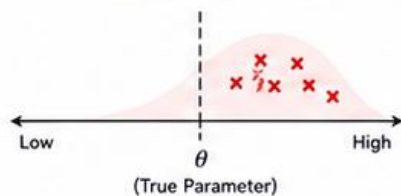


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